

Engineering Trivia — Teacher Guide

Should Have Known That · number sense, taught by a slider

WHAT THIS TEACHES

Forty verified numbers an engineer is expected to carry without looking up — **8 each** across Statics & Mechanics, Materials, Electrical, Fluids & Thermo, and Everyday Engineering. Students never type an answer; they **drag a slider** and commit. That single design choice is the pedagogy: a slider forces a magnitude out of someone who would otherwise say "I don't know," and a committed wrong answer is what makes the correction stick. Every reveal ends with a one-line **"you should have known that because..."** anchor, so the number leaves with a reason attached rather than as trivia. Runs offline in any browser — open **index.html**, phone or projector, nothing to install.

HOW IT IS SCORED

Thirty-five questions use a **linear** slider scored on **percent error**. Five use a **logarithmic** slider, marked **LOG SCALE** on screen, scored on **how many times off** you are: copper resistivity, the speed of light, the current that stops a heart, the voltage you can feel as a static shock, and the energy in an AA battery. Those five span orders of magnitude, so percent error would be meaningless — getting the right power of ten is the skill being tested.

BAND	LINEAR	LOG	PTS
■ Bullseye	within 5%	1.26×	100
■ Close	within 15%	2.1×	75
■ Warm	within 35%	5×	40
■ Cold	within 60%	16×	10
■ Way Off	past that	past that	0

Gold bands mean the student had the number, blue means they didn't, red means they weren't in the neighborhood. Thresholds live in the **SCORING** object at the bottom of **index.html** if you want a harsher or gentler curve.

20-MINUTE SESSION PLAN

- 0–2 min** Project the game (⌘ → **Presentation mode** for projector-sized type). Pick one category and **5 questions**. Play the first one yourself, out loud, and be wrong on purpose — it sets the tone that guessing is the assignment.
- 2–5 min** **Poll before every lock-in.** Hands up for higher, hands up for lower, then press LOCK IT IN. Public commitment before the reveal is the whole mechanism; a guess nobody said out loud costs nothing to be wrong about.
- 5–14 min** Switch to **teams** (2–4, pass-and-play on one device, or one device per team) and run **10 questions** each. Read the "because..." line aloud after every reveal — do not narrate over the animation, it only takes about two seconds.
- 14–18 min** Debrief from the end screen's **"Should have known that"** card, which names the game's biggest miss. Ask that team to say what number they *thought* it was and why. The wrong mental model is more useful to the room than the right answer.
- 18–20 min** Replay one category — **PLAY AGAIN** from the end screen, or ⌘ → **Reset** for a clean start screen. Scores jump, and that jump is the point: one exposure with a reason attached is most of what it takes to keep a number.

SOLO VS TEAM MODE

Solo is the homework and self-quiz mode: one player, a final score out of 100 per question, and the same two end cards. **Teams** (2–4) is the classroom mode. Name the teams on the start screen — it costs twenty seconds and buys you the hour. The game then hands the device around: a turn card names who is up, a strip across the top carries the running score, and every team answers the same number of questions.

The pool is split evenly, so the question count per team is capped at $\text{pool} \div \text{teams}$. With all five categories on and four teams, that is $40 \div 4 = \mathbf{10 \text{ questions each}}$. The note under the start button tells you the real number before you commit — if you switch categories off, watch it shrink.

One device or many? Pass-and-play on a single phone keeps the room together and is the intended mode. If each team has its own device, run the same category and length on all of them and compare end screens — but the question order is shuffled per device, so the scores are comparable only in aggregate, not question by question.

DISCUSSION PROMPTS

- 62.4 lb/ft³ and 0.433 psi/ft are the same fact.** Both are in the bank. Ask the room to derive one from the other ($62.4 \div 144 \text{ in}^2/\text{ft}^2$). One number they already had was two numbers they thought they didn't.
- Which number is the dangerous one** — 100 mA across the chest, or the 3,000 V it takes to feel a static shock? A doorknob zap is 5–25 kV and harmless. It is the current, not the voltage, that kills. Say plainly that 100 mA is a rule of thumb and not a safe/unsafe line: IEC 60479-1 puts fibrillation risk on a curve over current, duration and path, and well under 100 mA is already dangerous.
- Why not build everything out of aluminum?** Young's modulus 200 vs 70 GPa; density 7,850 vs 2,700 kg/m³. Both ratios are about 2.9, so stiffness-to-weight is nearly identical — the real choice is cost, buckling and joining.
- What if steel and concrete expanded at different rates?** They both run near 12 $\mu\text{m}/\text{m}\cdot^\circ\text{C}$. Every summer would crack the building. Reinforced concrete works on a coincidence.
- Did the steel melt?** Carbon steel melts near 1,500 °C, but it softens enough to fail far below that. Good for separating "melted" from "failed" in students' vocabulary.
- A36 is named after its own answer** — the 36 is the yield strength in ksi ($\approx 250 \text{ MPa}$). How many other "memorized" numbers are printed on the spec sheet name?
- One gallon of gasoline is 33.7 kWh** — roughly a full day of an average US home's electricity. Best single number in the bank for an energy conversation.
- Why did the log questions go so badly?** Because they are not recall questions. Tell the class this in advance; it lowers the cost of guessing and makes the failure informative instead of embarrassing.

Reading a Way Off: a Way Off on a **linear** question is a fact the student does not have. A Way Off on a **log** question is a bigger problem — they do not know what *size* the answer is. Say the difference out loud; it is the single most transferable idea in the game.

EDITING THE QUESTION BANK

Every question is one object in a commented array at the top of `index.html` — search for **QUESTION BANK**. The fields are `id`, `category`, `prompt`, `units`, `value` (the true answer), `min` / `max` (slider endpoints), `scale` (`"linear"` or `"log"`) and `factoid` (the "because..." line). Edit in place with any text editor; the game itself has no build step.

- New categories appear on the start screen and get a color **automatically** — just type a new `category` string.
- Keep the true value **off-center** in `[min, max]`, and keep the range wide enough that the endpoints don't give the answer away.
- Use `"log"` whenever plausible student guesses span orders of magnitude. Both `min` and `max` must be greater than zero on a log slider.
- Always write the `factoid`. A number without a reason is trivia, and trivia does not survive the week.

Murray State University · Bryant Harrison · Engineering Trivia (Should Have Known That) is an offline single-file web game — open `index.html` to play, `teacher-guide.html` to reprint this page. 40 questions, 8 per category; 5 logarithmic sliders; bands at 5/15/35/60% linear and 1.26/2.1/5/16× log.

One region of `index.html` is generated: the presenter notes, between the `__GUIDE__` markers. They are this guide, injected by `tools/guide-sync.js`, so the on-screen notes and the printed page cannot disagree. Edit `teacher-guide.html`, then run `node tools/guide-sync.js` and `node tools/pdf.mjs`.

CLASSROOM LOGISTICS

- **Nothing to install and nothing saved.** One file, no network, no accounts, no stored state — and **⌘ → Reset** puts the game back to a fresh load between class periods without reloading the page.
- **⌘ Settings** holds **Open Presenter Notes** (this guide, on screen), **Presentation mode** and **Reset**. **?** is the **Guide** button: the same overlay opens on every load, so students who arrive late can read the rules themselves, and **?** reopens it any time.
- **Keys: Enter** or **Space** locks in a guess and advances the reveal. **Esc** closes any popup. Everything else is touch.
- The reveal runs about **two seconds**: the truth marker slides out from under the guess, the band lands, then the "because..." line fades in and NEXT appears. Budget roughly 30–45 seconds per question with discussion.